Pinching-Antenna Systems (PASS): Paving the Way for 6G ISAC Innovations

Yunhui Qin, Yaru Fu, Yan Liang, Difei Cao, and Haijun Zhang

Abstract

Pinching Antenna SyStems (PASS) represent a groundbreaking leap in antenna technology, offering a flexible, adaptive, and cost-effective solution for next-generation integrated sensing and communication (ISAC) systems. With its unique ability to establish robust line-of-sight links using low-cost dielectric materials positioned arbitrarily along a dielectric waveguide, PASS represents a significant step forward in pursuing higher performance and efficiency for 6G networks. In this article, we embark on an exploration of PASS-enhanced ISAC systems, beginning with an overview of their significant potential across a range of cutting-edge applications. We then examine the core design challenges, including channel estimation, antenna positioning optimization, power allocation, and beamforming, while presenting innovative solutions to address these complexities. To validate these concepts, we showcase two compelling case studies that highlight the notable advantages of PASS. These examples illustrate how the adaptive antenna positioning and dynamic transmit power allocation corroboratively enhance both communication and sensing capabilities. Looking ahead, we chart a visionary roadmap for future research, identifying key directions to unlock the full potential of PASS in ISAC systems and catalyze transformative 6G innovations.

Introduction

Integrated sensing and communication (ISAC) has emerged as a groundbreaking technology that plays a pivotal role in shaping the sixth-generation (6G) communication systems [1]. It can incorporate wireless sensing and communication functions on a platform that leverages shared resources such as hardware, spectrum, and energy, thus improving operational efficiency, reducing costs, and promoting sustainability [2]. This capability addresses the extensive application demands of future intelligent scenarios such as intelligent driving, smart home, industrial internet of things (IIoT), and other domains [3].

The efficiency of shared resources utilization and the adaptability to dynamic environments present major challenges for ISAC systems, especially in their antenna systems. On the one hand, the sharing of resources between sensing and communication functions requires the use of highly directional and flexible antennas to mitigate signal interference and improve resource efficiency. On the other hand, in dynamic and complex environments, antennas must be capable of dynamically adapting to environmental variations, ensuring high communication quality and sensing accuracy. However, traditional fixed-position antenna designs often struggle to adapt to these diverse requirements.

In recent years, flexible antenna systems, such as reconfigurable intelligent surfaces (RIS) [4], intelligent reflecting surfaces (IRS) [5], and movable antennas (MA) [6], have attracted considerable attention. Due to their ability to dynamically reconstruct the wireless channel, flexible antenna systems can significantly enhance ISAC performance compared to traditional fixed-position antenna systems [7]. However, in most existing flexible antennas, the variation range of antenna positions is generally limited to the wavelength scale, constraining their ability to counteract large-scale path loss. In addition, the high costs of many existing flexible antennas also restrict their applications in real-world scenarios. The emergence of Pinching Antenna SyStem (PASS) overcomes these limitations, as it can create robust line-of-sight (LoS) links or enhance existing transceiver channels by using low-cost dielectric materials placed at arbitrary positions on a dielectric waveguide [8]. Table 1 compares the performance of PASS with other technologies based on existing studies [9–12]. It is worth noting that these prior works primarily focus on the optimization perspective and the performance trade-off between communication and sensing. However, numerous open questions remain unexplored, leaving significant opportunities for further research and innovation.

Considering the fertile opportunities and promising applications of PASS in ISAC towards 6G, this article aims to provide an overview of the fundamentals for PASS-enhanced ISAC as well as the main challenges in their implementation. To this end, we first discuss a comprehensive system framework and promising applications for PASS-enhanced ISAC systems. Then, we provide their core design challenges and potential solutions, including

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channel estimation, position optimization, power allocation, and flexible beamforming design. In addition, we present two compelling case studies that underscore the unique advantages of PASS, demonstrating how the adaptive antenna positioning and dynamic transmit power allocation corroboratively enhance both communication and sensing capabilities. Building on this, we outline a visionary research roadmap for future research, identifying critical pathways to fully harness PASS potential in ISAC systems and drive significant innovations for 6G development.

Framework and Promising Applications

In this section, we present a comprehensive system framework for PASS-enhanced ISAC networks and discuss its potential across a wide range of applications. Conventional ISAC integrates wireless sensing and communication functionalities into one system while sharing spatial, temporal, and frequency resources. PASS, as an innovative flexible antenna system, can further enhance ISAC by establishing robust LoS links. As shown in Fig. 1, a pinching antenna is the dielectric waveguide that works as a leaky-wave antenna by attaching other small dielectric particles on it. Therefore, PASS enables on-demand selection of radiation points along the waveguide via attached antenna for flexible communication area formation, while instantly terminating radiation and eliminating the communication area upon antenna release. Meanwhile, PASS can also receive signals at their radiation points and are easily installed by attaching separate dielectric components along the waveguide, thus possessing the potential to capture sensing echo signals [13]. In future 6G networks, the application scenarios of PASS-enhanced ISAC technology involve various domains such as intelligent driving, IIoT, smart home, extended reality (XR), etc., as shown in Fig. 1.

Intelligent driving: Intelligent driving requires highly reliable sensing and low-latency communication. Conventional ISAC systems struggle under adverse weather, network congestion, and security threats in hotspots. The PASS framework directly addresses these limitations by enhancing sensing accuracy, reducing communication latency, and establishing more robust and secure links, thereby enabling safer autonomous operations.

IIoT: Future industrial system will integrate sensing, control, and communication through intelligent automation to minimize human intervention. In such context, massive machine-type devices require ultra-reliable, low-latency connections to access points for real-time data exchange, posing greater challenges for ISAC networks. Due to the dynamic or low-mobility deployment of machine-type devices, conventional quasi-static wireless environments in ISAC face increasing demands for dynamic adaptability and reliability to enable full automation, intelligence, and seamless operation across IIoT workflows. PASS can establish flexible LoS links and improve wireless channel quality, thereby enhancing sensing adaptability and communication reliability by reducing both error rates and latency.

Smart home: Future smart homes will need wireless connections and data sharing between devices such as indoor sensors, domestic robots, and household appliances. These devices typically operate simultaneously to fulfill diverse smart service demands. As the number of connected devices grows, the computational and spectral resources available to each device become increasingly scarce. Most devices remain in fixed locations or move slowly. When wireless signals between devices and access points weaken, traditional ISAC adjustments often fail to compensate for these signal degradations cost-effectively. PASS, pre-installed around such areas, provides targeted signal enhancement adaptively, improving link reli-

Feature	PASS	IRS/RIS	MA
Active/Passive	Passive	Passive	Active
Flexibility	High	Low	Medium
Cost	Low	Medium	High
Scalable implementation	Easiest	Hard	Easily
Response speed	Fast	Extremely fast	Medium
Power consumption	Low	High	Medium

TABLE 1. Comparisons of PASS and other technologies.

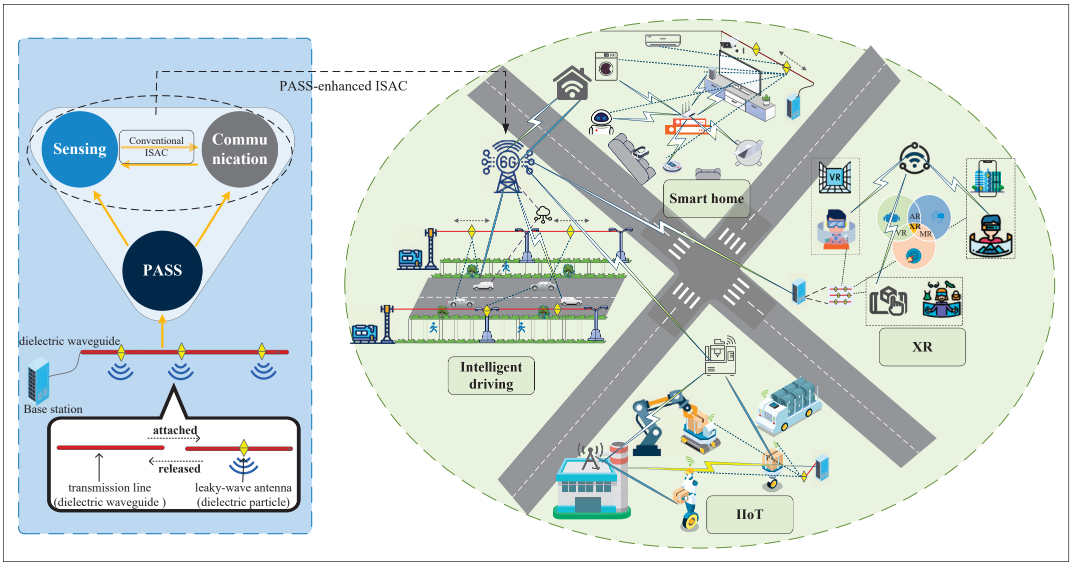

FIGURE 1. General system framework and potential applications of PASS-enhanced ISAC systems.

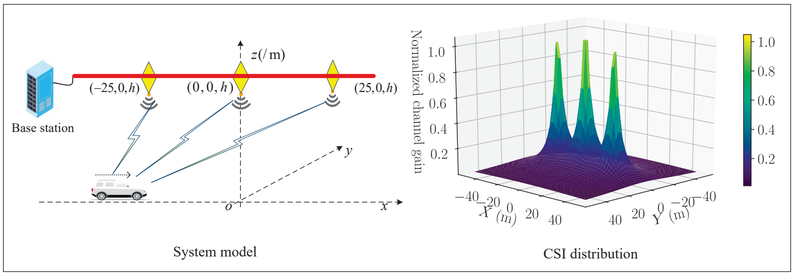

FIGURE 2. Illustration of a PASS scenario.

ability without requiring additional antennas.

XR: XR integrates virtual reality (VR), augmented reality (AR), and mixed reality (MR) through ultra-reliable, low-latency links for immersive experiences, which are challenged by dense device connectivity and diverse service requirements. Thereinto, sensing system performs real-time spatial mapping and user tracking, and communication infrastructure achieves low-latency and ultra-reliable data transmission. However, massive device connections, diverse user services, and stringent requirements for ultra-reliable and low-latency communication pose great challenges for conventional ISAC systems. The PASS framework effectively establishes robust line-of-sight links, reducing latency and alleviating network congestion to ensure a high-quality user experience.

Challenges and Potential Solutions

In this section, we examine the core design challenges, including channel estimation, antenna positioning optimization, power allocation, and beamforming, while presenting innovative solutions to address these complexities.

Channel Estimation for PASS-Enhanced ISAC

In PASS-enhanced ISAC systems, both user service and target sensing functionalities can be realized through LoS links enabled by pinching antennas. It is critical to acquire accurate channel state information (CSI) at the arbitrary positions and numbers of pinching antennas to ensure communication and sensing performance.

Figure 2 shows an example of the CSI characteristics of a PASS, which comprises a single waveguide with multiple pinching antennas. Specifically, multiple pinching antennas are uniformly deployed on the waveguide with fixed height h, and a vehicle moves at a constant velocity directly below the waveguide. Due to the propagation along the waveguide, the signal radiated from one antenna exhibits a fixed phase shift relative to the signal radiated from an adjacent antenna. The CSI in the vehicular receiver can be derived using the spherical wave propagation model, which is mainly determined by the relative positions of the antennas and the vehicular receiver [8]. Figure 2 provides a three-dimensional (3D) visualization of the CSI distribution.

It can be observed that the normalized channel gain decreases with increasing distance to the antennas, and consequently increases monotonically as the vehicle approaches the antennas. When the vehicle reaches the coordinate origin, the channel gain reaches its maximum value. In conventional ISAC systems, reinforcement learning (RL)-driven methods can be employed for CSI estimation. However, in PASS, the CSI of each wireless link may experience rapid fluctuations due to antenna displacement, since the antennas can be flexibly deployed to establish LoS links and exploit near-field advantages. This dynamic environment poses challenges even for RL-based channel estimation. To solve these challenges, knowledge-guided learning methods may be more effective in accelerating discovery of optimal CSI policies.

Antenna Position Optimization for PASS-Enhanced ISAC

Pinching antennas can be flexibly deployed at arbitrary positions on waveguides, while increasing their number incurs almost no additional cost. However, existing flexible antenna systems usually suffer from limited configuration flexibility, where adding or removing antennas is neither straightforward nor cost-effective. Many flexible antennas can only move within a few wavelengths, which has negligible impact on large-scale path loss. For example, if the LoS link is blocked, moving antennas by a few wavelengths often fails to restore it, and this limitation becomes more severe at high carrier frequencies (i.e., shorter wavelengths). In contrast, pinching antennas can move over larger distances, unrestricted by wavelength limitations. This offers the potential to combat large-scale path loss issues, such as maneuvering around obstacles to restore LoS links or positioning closer to targets/users, which would be impossible for other feasible antennas.

In dynamic scenarios caused by the movement of communication users or sensing targets, optimizing the positions of pinching antennas becomes more challenging. The exhaustive search over all feasible pinching antenna positions in the target region would ensure the optimal solutions. However, this method incurs high time and computational costs. RL methods may serve as effective solutions by providing offline trained policies that directly determine optimal configurations, thereby bypassing the need for real-time CSI estimation.

Power Allocation for PASS-Enhanced ISAC

In PASS-Enhanced ISAC systems, the allocation of transmit power refers to determining the power distribution between the sensing and communication functions. In considering waveguide power consumption scenarios, this should be included as a constraint. Power allocation enables joint sensing and communication optimization, improves energy efficiency, mitigates mutual interference, and adapts to dynamic requirements. Without efficient allocation, these systems suffer from unreliable sensing and poor spectrum utilization. As technology evolves, intelligent power allocation will remain essential to achieve increased safety, higher data rates, and improved energy sustainability in ISAC systems.

Particularly in non-orthogonal multiple access (NOMA)-ISAC system, the integration of PASS-enhanced adaptive LoS configuration with NOMA's power-domain multiplexing establishes a robust framework for next-generation wireless networks, laying the foundation for high-performance 6G communications. Therefore, power allocation serves as a critical enabler for optimizing multi-dimensional performance metrics, including energy efficiency, spectral efficiency, interference suppression, and sensing accuracy [11].

Flexible Beamforming Design for PASS-Enhanced ISAC

Dual-functional ISAC pinching beamforming design is facing critical challenges, which involve communications-centric, sensing-centric, and Pareto-optimal designs. Unlike existing flexible antenna systems, PASS achieves highly flexible beamforming via dynamic positioning combined with adaptive weights. Specifically, PASS enables each antenna to be independently activated or deactivated at any point along the waveguide, which facilitates dynamic reconfiguration of the antenna array, much like attaching or removing clothespins on clotheslines.

The optimal pinching beamforming strategy in existing studies aims to position pinching antennas as close as possible to the communication user or sensing target, while satisfying the minimum inter-antenna distance constraint. In time-varying scenarios involving mobile users or targets, conventional approaches struggle to meet the real-time updating requirements for beamforming. This poses a critical challenge for finding optimal solutions. Machine learning and RL techniques can offer promising solutions for these dynamic optimization challenges.

Case Studies

In this section, we present two case studies of PASS-enhanced ISAC systems, addressing the challenges introduced by the dynamic nature of channels. These case studies demonstrate how RL-driven solvers can effectively address the antenna positioning and power allocation problems, providing practical solutions to improve communication and sensing performance.

PASS-Enhanced OMA ISAC System

System Model: We consider an ISAC system consisting of a single waveguide with multiple pinching antennas, multiple mobile users (MUs), and sensing targets (STs). The single waveguide with multiple pinching antennas is integrated into a base station to establish LoS transceiver links, and pinching antennas are equipped on the same waveguide, meaning that the signal transmitted by one pinching antenna is a phase-shifted version of that transmitted by another. Time division multiple access (TDMA), an orthogonal multiple access (OMA) scheme, is utilized to serve single-antenna MUs and STs. Moreover, the wireless channels between MUs and pinching antennas are modeled on the basis of spherical wave channel model. Both the positions of pinching antenna and the transmission power significantly affect the communication and sensing performance.

In this case study, our objective is to maximize the total communication data rate while meeting the target sensing signal-to-noise ratio (SNR) and the system energy constraint. However, the optimization problem becomes complicated due to MUs' mobility, antenna positioning variations, and channel condition dynamics. Considering the inherent properties of the optimization problem, we propose an RL-based solution. Specifically, we define the state, action, and reward function as follows.

- *State*: the state mainly includes the locations of all pinching antennas, MUs and the STs, along with the total energy.
- *Action*: the action is defined as the variations in the locations of the pinching antennas, the locations of MUs, and the power allocation.
- *Reward*: the reward function is defined as the weighted total data rate and the sensing SNR.

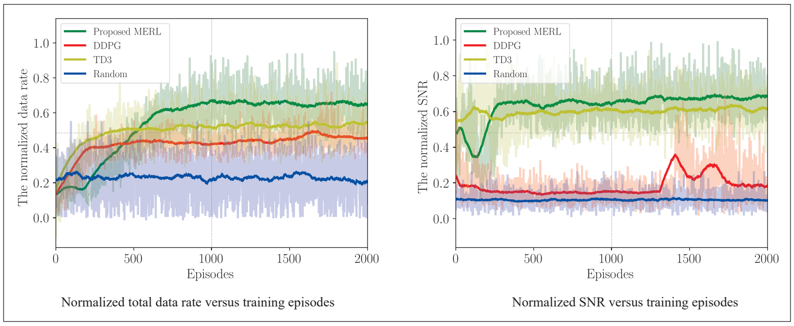

FIGURE 3. PASS-enhanced OMA ISAC system.

With the foregoing definitions, we elaborate on our proposed maximum entropy-based RL (MERL) solution [9]. This MERL focuses on maximizing cumulative reward and policy entropy, which ensures an effective balance between exploration and exploitation, resulting in enhanced robustness.

The detailed training process consists of two main phases: environmental interaction and network update. In the environmental interaction phase, the agent interacts with the environment, observes the current network state, selected action, immediate reward, and the next state, and then stores these experience tuples (state, action, reward, next state) in the experience replay buffer. During the network update phase, a mini-batch of tuples is randomly drawn from the buffer to update the actor and critic networks while optimizing policy entropy. Through this iterative process, the agent subsequently optimizes its policy and enhances overall system performance.

Numerical Results: We evaluate the performance of our proposed MERL scheme by numerical simulations. MUs and STs are randomly distributed within a 150 m × 150 m square area. The number of pinching antennas is 3, the number of MUs is 6, and the number of STs is 1. Other default settings for the environment and algorithms typically follow the parameters in [9]. For performance comparison, we utilize the following benchmark schemes:

- DDPG (deep deterministic policy gradient): This scheme is an off-policy RL algorithm, utilizing a deterministic policy with exploration noise.
- TD3 (twin delayed deep deterministic policy gradient): This scheme is an enhanced version of DDPG, employing two independent critic networks to mitigate overestimation and implementing delayed updates for the actor network to ensure stability.
- Random scheme: This scheme requires random values for the pinching antenna positions and the transmit power, each within predefined regions.

The normalized total data rate and the sensing SNR for our PASS-enhanced OMA ISAC system are shown in Fig. 3. It can be observed that the proposed MERL algorithm achieves a higher data rate compared to other benchmarks. Specifically, the proposed algorithm has improved the normalized total data rate by 20.3% compared to the TD3 algorithm and by 44.4% compared to the DDPG

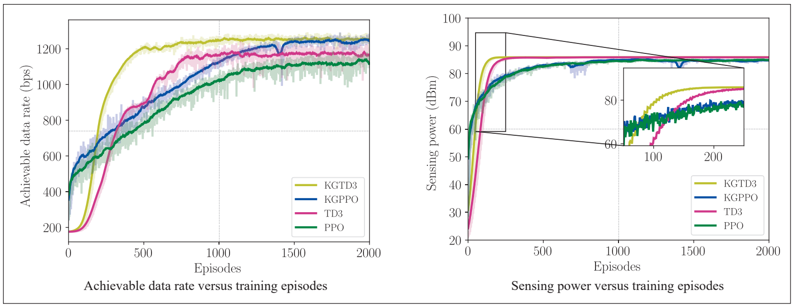

FIGURE 4. PASS-enhanced NOMA ISAC system.

algorithm. The proposed MERL algorithm achieves better normalized sensing SNR and stability compared to other benchmarks. It improves the normalized sensing SNR by 16.7% over TD3 and far exceeds that of the DDPG algorithm. Additionally, The proposed MERL shows non-increasing behavior initially, which stems from the inherent environmental stochasticity and the algorithm's deliberate exploration mechanism. In summary, guaranteeing a global optimum is intractable for the complex optimization problem in this case study. Nevertheless, our proposed method demonstrates superior performance over the baselines in terms of data rate and sensing SNR.

PASS-Enhanced NOMA ISAC System

System Model: NOMA scheme is used to serve single-antenna MUs and STs. Without loss of generality, all MUs are distributed into multiple user clusters. The user clustering mechanism defines the relationship between the pinching antennas and MUs, grouping the MUs into clusters where each cluster is associated with a pinching antenna. Besides, the MUs of each cluster are decoded using successive interference cancellation (SIC). Furthermore, the sensing power is determined by the channel vector between the STs and the pinching antennas and the covariance matrix of the transmitted signal.

This case study aims to maximize the communication data rate and target sensing power. However, jointly maximizing the communication and sensing performance by optimizing user clustering, inter-cluster transmit power allocation, and intra-cluster power allocation factors is challenging. Firstly, the problem is inherently complex due to its non-convex structure and the hybrid-variable nature of combining continuous and discrete parameters. Secondly, user clustering decisions are coupled with both intra-cluster and inter-cluster power allocation, creating nested dependencies that traditional optimization methods struggle to resolve. Given the aforementioned challenges, we propose RL-based solutions. Specifically, we define the state, action, and reward function as follows.

- *State*: the state mainly includes the locations of all pinching antennas, MUs and the STs, along with the total energy.
- *Action*: the action is defined as variations in the locations of the MUs, the locations of the STs, the allocation of power intra/inter-cluster, and the consumption of energy.
- *Reward*: the reward function is defined as the weighted achievable data rate and the sensing power.

Building on the foregoing definitions and the channel gain analysis (Fig. 2), we propose knowledge-guided RL algorithms that fuse channel gain-driven clustering principles with self-evolving optimization capabilities. Specifically, MUs are sorted by descending channel gain and clustered using gain-index pairing (e.g., pairs 1-4, 2-5, 3-6). Intra-cluster SIC decoding is then performed following the gain order. Finally, RL jointly optimizes intra-cluster and inter-cluster power allocation to maximize the data rate and sensing power. The proposed knowledge-guided RL algorithms include knowledge-guided proximal policy optimization (KGPPO) and knowledge-guided TD3 (KGTD3).

Numerical Results: The proposed knowledge-guided RL algorithms are evaluated by numerical simulations. Communication MUs and STs are randomly distributed within a 100 m × 100 m square area. We set the number of pinching antennas as 3, the number of MUs as 6, and the number of STs as 2. Other default settings for the environment and algorithms follow the parameters in [9]. For performance comparison, we use the following benchmark schemes.

- PPO: This scheme is an on-policy RL algorithm that trains policies by optimizing a clipped objective function, limiting policy updates to ensure stable learning.
- TD3: This scheme is an off-policy RL algorithm that improves stability over DDPG by using twin critic networks, delayed policy updates, and target policy smoothing.

The achievable data rate and sensing power for our PASS-enhanced NOMA ISAC system are shown in Fig. 4. The proposed algorithms achieve both higher data rates and faster convergence compared to the benchmark schemes. Specifically, the proposed KGTD3 approach demonstrates a significantly accelerated convergence during the initial training phase compared to TD3, while achieving 7% improvement in the average data rate. Compared to the baseline PPO, the proposed KGPPO exhibits faster convergence and 11% higher average data rate. Moreover, Fig. 4 shows that the proposed KGTD3 achieves better sensing performance during the initial training phase compared to its benchmark. Therefore, we infer that in our considered scenarios, prior knowledge of clustering principles effectively guides, accelerates, and improves the efficiency of the RL process.

Directions for Future Research

Research on the application of PASS in ISAC systems remains in its infancy, presenting numerous unresolved challenges and open research questions to be addressed in subsequent subsections.

Sensing Channel Modeling and Metrics

Sensing channel modeling for PASS-enhanced ISAC systems exhibits distinct characteristics compared to existing models, making its exploration critically necessary. Conventional ISAC systems define sensing as detecting targets and environmental features via wireless signals, whereas in the PASS-enhanced ISAC systems under consideration, sensing entails acquiring and interpreting multi-dimensional information spanning pinching antennas, networks, services, and targets. Two issues require further study: first, the impairment of sensing channels by multipath fading, shadowing, or interference; and second, the techniques for the reception of echo signals in such challenging environments.

The characterization of sensing metrics in PASS-enhanced ISAC systems presents unresolved challenges and constitutes a critical research topic, including sensing rate, sensing mutual information, the target accuracy, the probing power, interference management, and Cramér-Rao lower bound, among others. Furthermore, future research will include a theoretical analysis of sensing performance, linking the PASS geometry to ranging and estimation bounds, as well as collaboration with the industrial sector to build prototypes.

Uplink Reception Design for PASS-Enhanced ISAC

The design of uplink reception presents a highly challenging problem, particularly in ISAC systems, where the reception of echo signals is crucial for near-field sensing, localization, and performance analysis. Simultaneous transmission and reception using pinching antennas within the same waveguide is highly complex and often impractical. Therefore, separating the transmit and receive waveguides presents an effective solution. Furthermore, this approach can help mitigate other practical challenges that degrade performance, such as dielectric loss, mechanical/thermal instability at pinching points, control latency, and mutual coupling effects.

Interference Management for PASS-Enhanced ISAC

Inter-waveguide interference in multi-waveguide PASS systems is a critical and complicated issue, particularly in multi-user scenarios like NOMA, making it challenging to distinguish the strong and weak ones, as their received signal strength is primarily influenced by their respective locations, antenna positions, and the transmit power of the other users.

It is urgent to explore several techniques that hold potential for mitigating this interference, including inter-waveguide spatial multiplexing, interference suppression aided by other flexible antennas, and advanced multi-user detection schemes. Specifically, multiple waveguides offer spatial degrees of freedom that enable signal precoding, mitigating interference at the source. As a complementary approach, technologies like IRS can shape the propagation environment to nullify interference at unintended receivers. Finally, advanced multi-user detection algorithms can be developed to decode desired signals efficiently despite interference.

Artificial Intelligence for PASS-Enhanced ISAC

Artificial intelligence (AI) plays a pivotal role in facilitating the implementation of PASS-enhanced ISAC systems by enabling real-time optimization of joint communication and sensing performance. The joint optimization problems for communication and sensing may involve hybrid continuous-discrete variables with nonconvex and nonlinear properties. This complexity arises from three factors: the imperfect CSI, the mobility of MUs and STs, and the phase shifts (including free-space and waveguide propagation phase shifts) determined by the locations of pinching antennas.

For scenarios with imperfect CSI, an efficient method would be to perform an exhaustive search over all feasible antenna positions in the target region and select the positions that maximize both communication and sensing performance. However, ensuring optimality requires a sufficiently large number of feasible antenna positions, which results in prohibitively high time and computational costs. To address these challenges, AI methods, such as RL and deep neural networks, may serve as effective solutions. For example, pinching antenna positions can be optimized without requiring perfect CSI. Instead, RL algorithms can provide offline trained decision policies to directly determine optimal configurations, bypassing the need for real-time CSI estimation. To address the prohibitive computational cost of exhaustive search, AI-driven approaches present a promising direction for future research. Besides, comparisons with other antenna technologies, such as RIS-based ISAC, under similar AI solutions are also important directions for future research.

Integrating PASS with Other Flexible Antennas for ISAC

In the highly dynamic environments of intelligent driving, the spatial distribution of vehicles exhibits rapid fluctuations, while the presence of significant large-scale fading effects caused by buildings, trees, and even moving objects poses critical challenges to conventional ISAC systems. To address these challenges, the integration of pinching antennas with other flexible antenna systems shows substantial potential. Distributed pinching antennas connected to base stations can exploit their effective and robust LoS capabilities to maintain stable links in traffic hotspots. Base stations equipped with pinching antennas can dynamically adjust both the position and direction of their beams, thereby fully leveraging spatial multiplexing gains. Therefore, studying the synergy effects among multiple flexible antenna systems is worthwhile, as it not only enhances the quality of signal coverage but also improves the overall sensing and communication performance in highly dynamic and complex driving environments.

Conclusion

In this article, we embarked on an exploration of PASS-enhanced ISAC systems, beginning with a comprehensive system framework and its potential across a wide range of applications. We then examined the core design challenges, including channel estimation, antenna positioning optimization, power allocation, and beamforming, while presenting innovative solutions to address these complexities. Through two compelling case studies, we highlighted the notable advantages of PASS-enhanced ISAC. These examples illustrated how the adaptive antenna positioning and dynamic transmit power allocation corroboratively enhance both communication and sensing capabilities. Consequently, based on these findings, we outlined future research directions to pave the way for transformative 6G applications and services.

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References

[1] H. Zhang *et al.*, "Integrated Sensing and Communication for 6G Holographic Digital Twins," *IEEE Wireless Commun.*, vol. 32, no. 2, 2025, pp. 104–12.

[2] H. Zhang *et al.*, "Joint Radar Sensing, Location, and Communication Resources Optimization in 6G Network," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 9, pp. 2369–2379, 2024.

[3] H. Zhang *et al.*, "USTB 6G: Key Technologies and Metaverse Applications," *IEEE Wireless Commun.*, vol. 30, no. 5, 2023, pp. 112–19.

[4] R. Liu *et al.*, "Integrated Sensing and Communication with Reconfigurable Intelligent Surfaces: Opportunities, Applications, and Future Directions," I*EEE Wireless Commun.*, vol. 30, no. 1, 2023, pp. 50–57.

[5] S. Xu *et al.*, "Intelligent Reflecting Surface Enabled Integrated Sensing, Communication and Computation," *IEEE Trans. Wireless Commun.*, vol. 23, no. 3, 2023, pp. 2212–25.

[6] L. Zhu, W. Ma, and R. Zhang, "Modeling and Performance Analysis for Movable Antenna Enabled Wireless Communications," *IEEE Trans. Wireless Commun.*, vol. 23, no. 6, 2023, pp. 6234–50.

[7] K. Wang, Z. Ding, and G. K. Karagiannidis, "Antenna Activation and Resource Allocation in Multi-Waveguide Pinching-Antenna Systems," *IEEE Trans. Wireless Commun.*, 2025, doi:10.1109/TWC.2025.3608068.

[8] Z. Ding, R. Schober, and H. V. Poor, "Flexible-Antenna Systems: A Pinching-Antenna Perspective," *IEEE Trans. Commun.*, vol. 73, no. 10, 2025, pp. 9236–53.

[9] Y. Qin, Y. Fu, and H. Zhang, "Joint Antenna Position and Transmit Power Optimization for Pinching Antenna-Assisted ISAC Systems," *IEEE Wireless Commun. Lett.*, vol. 14, no. 11, 2025, pp. 3535–39.

[10] C. Ouyang *et al.*, "Rate Region of ISAC for Pinching-Antenna Systems," arXiv preprint arXiv:2505.10179, 2025.

[11] Y. Liu *et al.*, "Pinching-Antenna Systems (PASS): Architecture Designs, Opportunities, and Outlook," *IEEE Commun. Mag.*, 2025. DOI: 10.1109/MCOM.001.2500037.

[12] C. Ouyang *et al.*, "ISAC Rate Region of Pinching-Antenna Systems," *2025 IEEE/CIC Int'l. Conf. Commun. in China (ICCC Workshops)*, Aug. 2025, pp. 1–6.

[13] H. O. Y. Suzuki and K. Kawai, "Pinching Antenna: Using a Dielectric Waveguide as an Antenna," *NTT DOCOMO Technical J*, vol. 23, no. 3, 2022, pp. 5–12.

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